

Exercise 4

1. SELECT A.Employee-ID, A.FirstName, ^A Last Name,
A department, A salary, B ProjectID,
B.ProjectName, B Budget, B Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

2. SELECT A.Employee-ID, A.FirstName, A.LastName,
A department, A salary, B ProjectID, B ProjectName,
B Budget, B status
FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial analysis	90000	Ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	Wilson	Sales	65000	105	Market expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

4

SELECT A.Employee_ID, A.FirstName, A.LastName,
A.department, A.Salary, B.ProjectID,
B.ProjectName, B.Budget, B.Status.

FROM Employees AS A.

FULL OUTER JOIN Projects AS B

ON A.EmployeeID = B.EmployeeID

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

6

7. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Salary > 70 000

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

8. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department = 'IT' OR 'Finance'

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9. SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE status = ~~'On going'~~ ~~'Not completed'~~ OR 'Pending.'

ProjectID	ProjectName	Budget	Status
102	Employee training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

10. SELECT ProjectID, Project Name, Budget, status
FROM Projects
WHERE Budget > 70000 AND status NOT 'completed'

Project ID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

11. SELECT EmployeeID, FirstName, LastName, City, Salary
FROM Employees
WHERE City = 'New York' OR 'Toronto'
ORDER BY Salary DESC

EmployeeID	FirstName	lastName	city	Salary
6	Michael	Clark	New york	80000
3	Bob	Johnson	Toronto	75000
1	John	Doe	New york	70000

12.

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SELECT EmployeeID, FirstName, lastName, Department, Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3.
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EmployeeID	FirstName	lastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

13.

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SELECT Department,
SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY department
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ORDER BY SUM(Salary) DESC
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Department	TotalSalary
IT	142000
HR	60000
Finance	155000
Sales	65000

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

14. SELECT city
 AVG(salary) AS AverageSalary
 FROM Employees
 GROUP BY city
 HAVING AVG(salary) > 65000

City	Average salary
New York	75000
Toronto	75000
London	72000

15. SELECT department,
 COUNT(EmployeeID) AS EmployeeCount
 GROUP BY department
 HAVING COUNT(EmployeeID) > 1

department	EmployeeCount
IT	2
Finance	2

16. SELECT status,
 COUNT(ProjectID) AS ProjectCount
 FROM Projects
 GROUP BY status
 HAVING COUNT(ProjectID) ≥ 2

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2


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17 SELECT A.EmployeeID,  
        A.FirstName  
        A.LastName  
        B.(Sum(Budget) AS TotalProjectBudget)  
FROM Employees AS A  
Inner Join Projects AS B  
ON A.EmployeeID = B.EmployeeID  
GROUP BY EmployeeID  
HAVING Sum(Budget) > 150000
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EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000